

Tongyao Zhang

Department of Psychological and Brain Sciences
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Education

2021 – 2026 Ph.D. in Psychology, Indiana University, United States

Concentration: Developmental Psychology and Cognitive Science

Advisory committee: Dr. Emily R. Fyfe (chair), Dr. Karin H. James, Dr. Linda B. Smith, Dr. Robert L. Goldstone

2016 – 2020 B.Sc. in Psychology, Beijing Normal University, China

Concentration: Cognitive and Developmental training,

Advisors: Dr. Qingfen Hu, Dr. Roi Cohen-Kadosh, Dr. Xinlin Zhou

Research Positions

July 2026 – present Postdoctoral Fellow, Indiana University, United States

[AI in Education Project](#) with [Terracotta](#), PI: Dr. Benjamin Motz

Honors and Awards

2025 Foundry10 Travel Award for the Mathematical Cognition and Learning Society, \$600

2024 Kempf Fellowship in Psychology, \$6,000

2022 – 2024 NIH T32 Training Grant in Integrative Developmental Process, Director: Dr. Linda B. Smith, \$25,000/year

2019 Second-Class Outstanding Undergraduate Scholarship, Beijing Normal University

2019 Fung Scholarship, University of Oxford (funded by Victor and William Fung Foundation), PI: Dr. Roi Cohen-Kadosh, £3,500

2018 Third-Class Outstanding Undergraduate Scholarship, Beijing Normal University

Publications

Borriello, G. A., **Zhang**, T., Wang, W., Peng, P., & Fyfe, E. R. A meta-analytic review of the associations between patterning and academic achievement. (In press in *Psychological Bulletin*)

- Zhang**, T., Borriello, G. A., James, K. H., & Fyfe, E. R. (2025). The role of patterning skill in cognitive development and learning: A critical review. *Developmental Review*, 76, 101202. <https://doi.org/10.1016/j.dr.2025.101202>
- Gök*, S., **Zhang***, T., Goldstone, R. L., & Fyfe, E. R. (2025). Multimedia effects in initial instruction and feedback on problem solving. *Journal of Educational Psychology*, 117(7), 1077–1094. <https://doi.org/10.1037/edu0000943> (*shared first authorship)
- Motz, B., Diekman, A., Goldstone, R., Green, D., Fyfe, E., Bernardini, S., ...**Zhang**, T., & Murphy, M. (2025). A field-initiated vision of research infrastructure for STEM education. *International Journal of STEM Education*, 12(1), 1–13. <https://doi.org/10.1186/s40594-025-00581-z>
- Zhang**, T., & Fyfe, E. R. (2024). High variability in learning materials benefits children’s pattern practice. *Journal of Experimental Child Psychology*, 239, 105829. <https://doi.org/10.1016/j.jecp.2023.105829>

Manuscripts in Progress

- Zhang**, T., & Fyfe, E. R. The effect of combined patterning and numeracy instruction on children’s mathematics learning. (Under revision)
- Zhang**, T., & Fyfe, E. R. Promoting abstract thinking with symbolic labels in pattern practice. (Under review)
- Zhang**, T., & Fyfe, E. R. Tuned to Focus on Structure: Connection between Patterning and Reasoning. (In preparation)

Talks and Conference Presentations

- Zhang**, T., Fyfe, E. R. (expected 2026, November). Patterning Practice Tunes Learners to Focus on Structures. Poster to be presented at the Psychonomic Society 67th Annual Meeting, San Diego, CA, United States.
- Zhang**, T., Gök, S., Gold, G., Carvalho, P. F., Fyfe, E. R. (2026, July). Effectiveness and Efficiency of Multimedia Worked Examples vs. Practice with Feedback on Learning Problem Solving. Poster virtually presented at the Cognitive Science Society, Rio de Janeiro, Brazil.
- Zhang**, T., Fyfe, E. R. (2026, April). “It’s always the same pattern”: Promoting abstract thinking with symbolic labels in pattern practice. Poster presented at the Biennial Meeting of the Cognitive Development Society (CDS), Montréal, Canada.
- Zhang**, T. (2026, February). From Patterns to Numbers: The Interplay of Cognitive Skills in Early Learning. Invited talk at the Brain and Cognitive Development Lab, University of Illinois Urbana-Champaign, United States.

- Zhang, T., Fyfe, E. R. (2025, June).** Tuned to focus on structure: Connection between patterning and reasoning. Poster presented at the Mathematical Cognition and Learning Society (MCLS), Hong Kong S.A.R., China.
- Zhang, T., Fyfe, E. R. (2025, May).** Counting with patterns: The effect of including patterns in numeracy instructions. Poster presented at the Society for Research in Child Development (SRCD), Minneapolis, MN, United States.
- Zhang, T., & Fyfe, E. R. (2024, June).** Malleability of various patterning skills in 5- and 6-year-old children. In Nicholas A. Vest & Chen Xueliang (Chairs), Pattern learning: Empirical research about interventions, parental beliefs, and links to mathematical competence in children [Symposium]. Mathematical Cognition and Learning Society (MCLS), Washington, DC, United States.
- Gök, S., **Zhang, T., Goldstone, R., Fyfe, E. R. (2024, April).** Multimedia materials are more effective in feedback than in main instruction. Poster presented at the American Educational Research Association (AERA), Philadelphia, PA, United States.
- Zhang, T., Gök, S., Fyfe, E. R., Goldstone, R. (2024, April).** The impact of instructional visualizations on students' drawings during mathematical problem solving. Poster presented at the American Educational Research Association (AERA), Philadelphia, PA, United States.
- Zhang, T. & Fyfe, E. R. (2024, March).** Not all pattern tasks are equal: Predicting children's numeracy skills from early patterning assessments. Poster presented at the Biennial Meeting of the Cognitive Development Society (CDS), Pasadena, CA, United States.
- Borriello, G. A., Peng, P., **Zhang, T., Fyfe, E. R. (2023, March).** A meta-analysis examining relations between patterning, cognition, and math and reading achievement. Poster presented at the Society for Research in Child Development (SRCD) 2023 Biennial Meeting, Salt Lake City, UT, United States.
- Zhang, T. & Fyfe, E. R., (2023, June).** The benefits of mixing it up: Variable materials influence children's pattern practice. Talk presented at the Midwest Cognitive Science Conference, Grand Rapids, MI, United States.
- Zhang, T. & Fyfe, E. R., (2022, March).** Variability in Pattern Practice. Poster presented at the Biennial Meeting of the Cognitive Development Society (CDS), Madison, WI, United States.

Journal Reviewing

Journal of Experimental Child Psychology
 Child Development (co-review)

Teaching

Spring 2027 (Planned)

Instructor of Record, Indiana University

The Origin and Development of Number Sense

Fall 2025

Instructor of Record, Indiana University

Lab in Developmental Psychology

Fall 2024 – Spring 2025

Associate Instructor, Indiana University

Lab in Developmental Psychology

Fall 2022 – Spring 2024

Associate Instructor and Guest Lecturer, Indiana University

Developmental Psychology

Fall 2021 – Spring 2026

Graduate Teaching Assistant, Indiana University

Developmental Psychology; Methods of Experimental Psychology; Lab in

Developmental Psychology; Infant Development

Undergraduate Mentoring

Summer 2026

Neharika Upadhyay

Fall 2025 – Spring 2026

Avery Refka, Ameer Abu-Salih

Fall 2025

Willow Burton

Fall 2024

Vigna Inturi

Spring 2024 – Spring 2025

Nicole Namack

Fall 2023 – Spring 2025

Ben Hanson

Fall 2023 – Summer 2024

Allyson Utz, Jillian Pullen

Spring 2022 – Summer 2023

Anna Settle

Fall 2022 – Spring 2023

Rutvi Shah

Spring 2022 – Summer 2022

Wenwen Du

Memberships

American Educational Research Association (AERA)

Cognitive Development Society (CDS)

Cognitive Science Society (CogSci)

Mathematical Cognition and Learning Society (MCLS)

Society for Research in Child Development (SRCD)

Skills

Programming: R, JsPsych, PsychoPy, MATLAB, C#

Language: Mandarin (Native), English (Fluent), Japanese (Intermediate) – JLPT N2